SOFC POC · 2026-11-04

Hot standby → cooling: the plant slope, and the twin that now uses it

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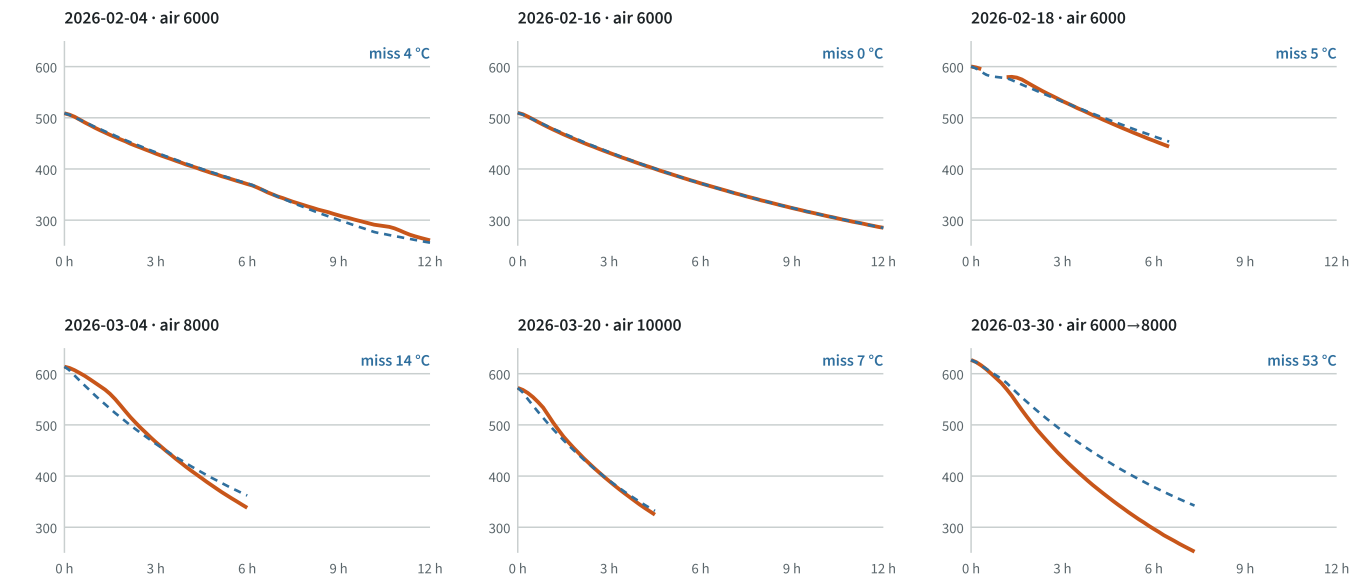
POC DATA · HOT STANDBY → AIR-FLOW COOLING

How fast the stack cools after hot standby

When the plant leaves hot standby and only air blows through, the stack (the fuel-cell block) cools at about **30 °C per hour at normal air** and **60–70 °C per hour at strong air**. It starts gently for the first 15 minutes, is fastest in the first hours, then slows down as it gets cooler.

The twin now uses this measured slope. Before, it used its own heat balance (a simple heat-in / heat-out sum), which missed the plant by 12 °C on average on these runs; now it misses by 5 °C.

The six times the plant did it



Stack temperature (°C, median of the stack thermocouples) for up to 12 h after hot standby ended. Each run is cut where the plant switched the air heater back on or changed stage. The twin gets the plant's air flow minute by minute. "Miss" = average gap in °C. 02-18 has a 1-h logging gap.

The slopes, run by run

Run	First 15 min °C/h	Fastest in 15–60 min	1–3 h	3–6 h	50 °C down after	100 °C down after
2026-02-04 · from 509 °C · air 6000	-20	-31	-25	-20	1.8 h	4.0 h

Run	First 15 min °C/h	Fastest in 15–60 min	1–3 h	3–6 h	50 °C down after	100 °C down after
2026-02-16 · from 510 °C · air 6000	-20	-31	-25	-20	1.8 h	4.0 h
2026-02-18 · from 600 °C · air 6000	-15	-8	-29	-26	2.4 h	4.2 h
2026-03-04 · from 614 °C · air 8000	-18	-38	-62	-43	1.4 h	2.2 h
2026-03-20 · from 572 °C · air 10000	-26	-69	-63	-44	1.0 h	1.6 h
2026-03-30 · from 627 °C · air 6000 → 8000 after 1 h **(only 14 min in hot standby, straight after part load)**	-30	-55	-74	-46	1.1 h	1.7 h

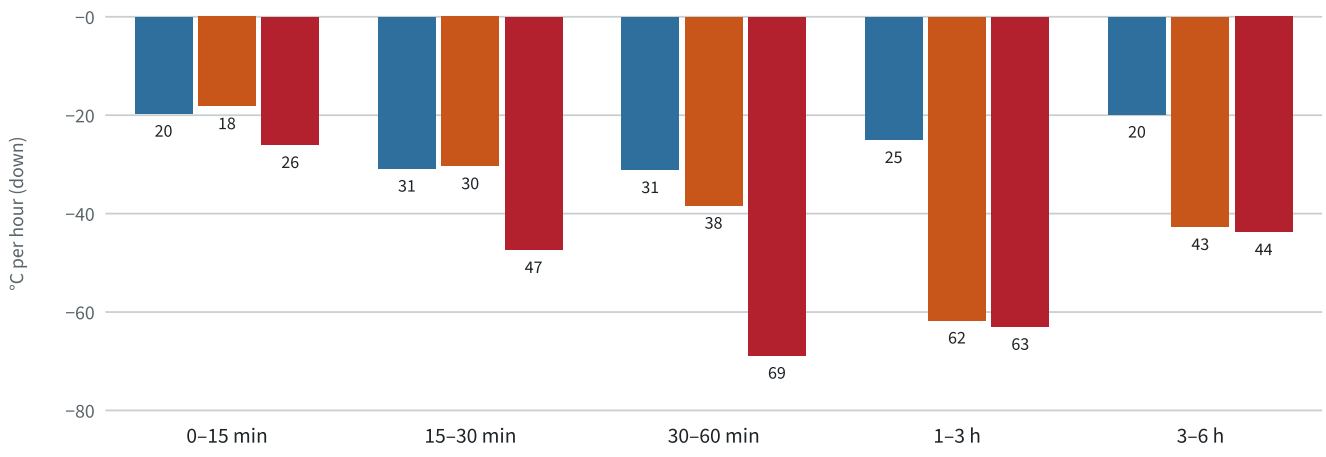
Minus sign = cooling. "Normal air" is 6000 ln/min (litres per minute at standard conditions); "strong air" is 8000–10000.

> **If you remember one thing: from hot standby (~510 °C) the plant drops 50 °C in about 1.8 h at normal air and 1–1.5 h at strong air; the twin now follows that.**

SHAPE OF THE CURVE

Gentle start, fast middle, slow tail

The cooling curve bends three times. In the first minutes the stack hardly moves. Within 15–30 minutes it reaches full speed. After that it slows down steadily, because a cooler stack has less heat left to give to the air.



Normal air 6000 (02-04, 02-16, from 509 °C) Strong air 8000 (03-04, from 614 °C)

Strong air 10000 (03-20, from 572 °C)

Average cooling speed of the plant stack in each time window after hot standby ended (straight-line fit per window). Taller bar = faster cooling. Strong air also starts slowly because in those runs the plant kept its air heater running for 16–51 min after leaving hot standby.

Why it starts slowly

- The air heater (EAPH, the electric heater on the incoming air) is ramped down, not switched off at once. In three of the six runs it kept heating for 16 to 51 minutes after hot standby ended.
- The hot box around the stack, the heaters and the recuperators (heat exchangers that pre-warm the incoming air with the exhaust) are still hot and give heat back at first.

Run	Time in hot standby before	Air heater off after
2026-02-04	80 h 14 min	0 min
2026-02-16	55 h 39 min	0 min

Run	Time in hot standby before	Air heater off after
2026-02-18	15 h 07 min	35 min
2026-03-04	6 h 02 min	51 min
2026-03-20	2 h 07 min	16 min
2026-03-30	0 h 14 min	0 min

The rule the twin now uses

Cooling speed depends on two things: how far the stack is above room temperature, and how much air blows. Fitted on the five clean runs (all except 03-30), this gives:

Stack temperature	air 6000	air 8000	air 9000	air 10000
600 °C	-38	-62	-75	-89
500 °C	-29	-46	-56	-67
400 °C	-20	-32	-39	-47
300 °C	-13	-20	-25	-29

°C per hour, once the first minutes have passed (the speed builds up over about 4 minutes). More air cools faster than in proportion: 9000 ln/min cools almost twice as fast as 6000.

FORMULA IN THE TWIN (APP.JS, HSCOOLRATE)

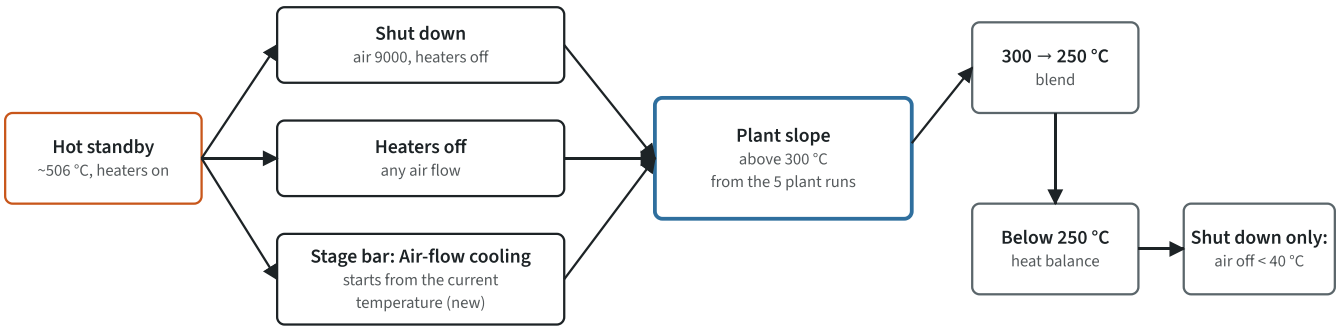
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slope = -0.0605 × (air ÷ 6000)^1.65 × u × √(u ÷ 475) × (1 - e^(-t ÷ 4.2 min)), u = stack - 25 °C
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> **If you remember one thing: about −29 °C/h at 506 °C with normal air, about −57 °C/h with 9000 air, and both halve by the time the stack is near 330 °C.**

DIGITAL TWIN · VERSION 25

What changed in the twin

Whenever the twin leaves hot standby into air-flow cooling, its stack now cools at the plant's measured slope instead of the twin's own heat-balance estimate.



Three ways out of hot standby into cooling all use the plant slope. The plant runs never went below about 260 °C, so from 300 °C down the twin hands back to its heat balance.

Replay of the six plant runs

Run	Twin before (miss)	Twin now (miss)
2026-02-04 · from 509 °C · air 6000	10.5 °C	4.5 °C
2026-02-16 · from 510 °C · air 6000	12.8 °C	0.5 °C
2026-02-18 · from 600 °C · air 6000	6.0 °C	5.1 °C
2026-03-04 · from 614 °C · air 8000	17.3 °C	14.2 °C
2026-03-20 · from 572 °C · air 10000	16.1 °C	6.6 °C
2026-03-30 · from 627 °C · air 6000 → 8000 after 1 h	66.4 °C	52.9 °C

Average over the five clean runs: **12.2 → 5.1 °C**. Fair test (each run left out of the fit in turn, then predicted): 10.8 °C, so the gain on new runs is smaller than on these five. The 03-30 run is still 53 °C off (see Still open).

From the twin's own hot standby (506 °C)

What you press	after 1 h	3 h	6 h	12 h
Heaters off, air 6000	485 → **479**	443 → **430**	387 → **369**	296 → **283**
Shut down button (air 9000)	463 → **455**	382 → **373**	287 → **287**	166 → **168**
Heaters off, air 10000	458 → **447**	367 → **354**	266 → **261**	144 → **142**
Plant 02-04, air 6000, from 509 °C	481	430	371	261

Stack °C, before → **now**. With normal air the twin now lands within 2 °C of the plant's 02-04 run at 1, 3 and 6 h.

New on the twin page

- Switch **Cooling after hot standby: Plant slope** / **Heat balance**, to compare the two.
- Clicking **Air-flow cooling** in the stage bar while in hot standby now starts a real cool-down from the current temperature. From any other stage it still jumps to 242 °C as before.

WHERE IT LIVES

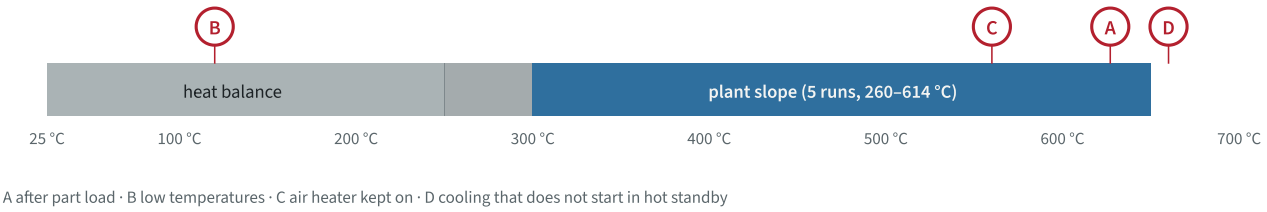
`twin_2d/app.js: hsCoolRate(), step(), jump() · twin_2d/index.html: switch + How it works · twin_2d/README.md`

> **If you remember one thing: the twin's cool-down out of hot standby now matches the plant within about 5 °C on the plant's own runs.**

DECISIONS FOR YOU

Still open

Four points the five plant runs cannot settle on their own.



A after part load · B low temperatures · C air heater kept on · D cooling that does not start in hot standby

Where the plant slope is based on data (blue) and where the twin still uses its heat balance (grey).

	Open point	What we see	Options
A	Cooling straight after part load	03-30: only 14 min in hot standby after part load, from 627 °C. Plant cooled much faster (−74 °C/h at 8000 air); the twin is still 53 °C off after a few hours.	Keep as is (one run only) · or add a "fresh from part load" boost once more such runs exist.
B	Below 300 °C	No hot-standby run went below ~260 °C. The plant's full cool-down after trip #8 (air 9000) reached 41 °C in 18 h; the twin's Shut down needs ~34 h to 40 °C.	Fit the low end on that trip run and let the plant slope run down to 40 °C · or keep the heat balance.
C	Air heater kept on after hot standby	In 3 of 6 runs the plant ramped its air heater down over 16–51 min; the twin switches it off at once and covers this with a 4-min start-up on average.	Keep the average start-up · or ramp the twin's air heater down like the plant.
D	Cooling that does not start in hot standby	After a trip purge the plant cooled slower at 6000 air (−9 to −25 °C/h). The plant slope is only used out of hot standby.	Keep it limited to hot standby · or fit a separate slope for after-trip cooling.

If you remember one thing: the plant slope is solid between ~300 and ~620 °C out of a settled hot standby; outside that it rests on one run or none.